Introduction to Statistical Investigations

Second Edition

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Chapter 1.4

Significance: How Strong is the Evidence?

1.4: What Impacts Strength of Evidence?

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Quizzes

- Study your quiz and figure out what the correct answers/processes are
- You can correct your quiz, from now until March 5th
- You can attempt up to two times
- You get to see your previous attempts, and a calculator.
- You'll have at most 25 minutes to do complete it.

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Example 1.4: Predicting Elections

- Do voters make judgments about candidates based on facial appearances?
- More specifically, can you predict an election by choosing the candidate whose face is more competent-looking?
- Participants were shown two candidates and asked who has the more competent-looking face.

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Who Has the More Competent Looking Face?

- 2004 Senate Candidates from Wisconsin

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Who Has the More Competent Looking Face?

- 2004 Senate Candidates from Wisconsin

Winner

Loser

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Example 1.4: Predicting Elections (1 of 4)

- They determined which face was the more competent for the 32 Senate races in 2004.
- What are the observational units?

- What is the variable measured?

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Example 1.4: Predicting Elections (1 of 4)

- They determined which face was the more competent for the 32 Senate races in 2004.
- What are the observational units?
  - The 32 Senate races
- What is the variable measured?
  - Whether the method predicted the winner correctly

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Example 1.4: Predicting Elections (2 of 4)

- Null hypothesis: The probability this method predicts the winner equals 0.50. ($H_0 : \pi = 0.50$)
- Alternative hypothesis: The probability this method predicts the winner is greater than 0.50. ($H_0 : \pi > 0.50$)
- This method correctly predicted 23 of 32 races, hence $\hat{p} = \frac{23}{32} \approx 0.719.$

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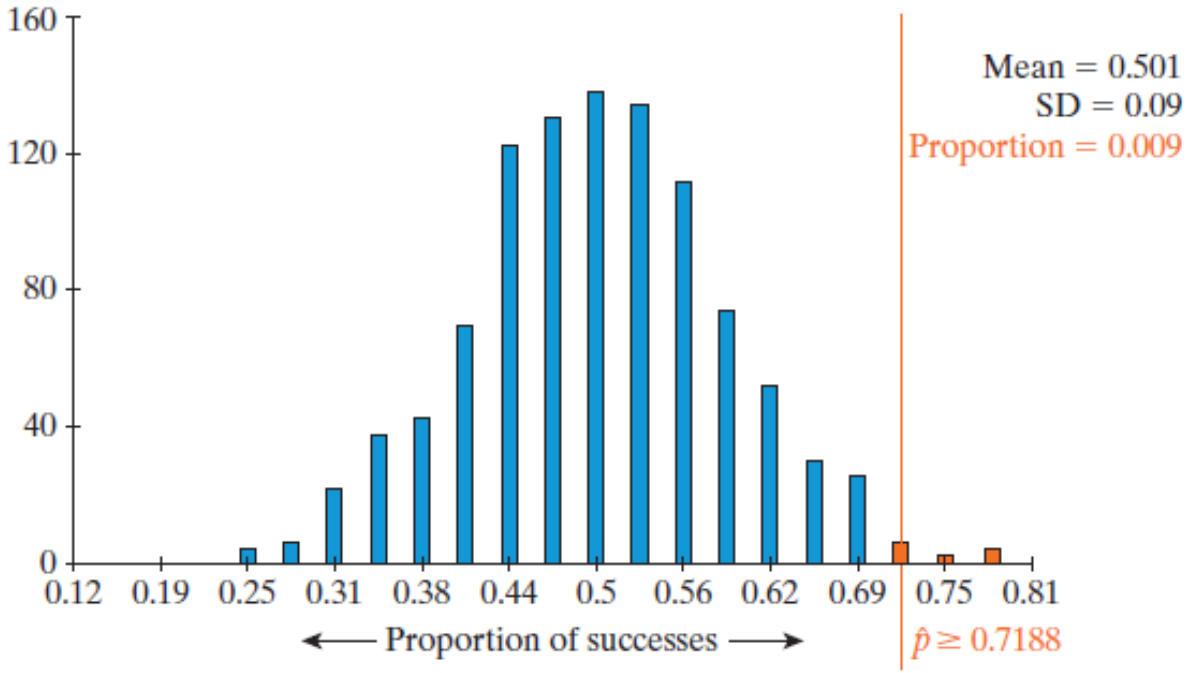
Example 1.4: Predicting Elections (3 of 4)

1,000 simulated sets of 32 races using the One Proportion applet.



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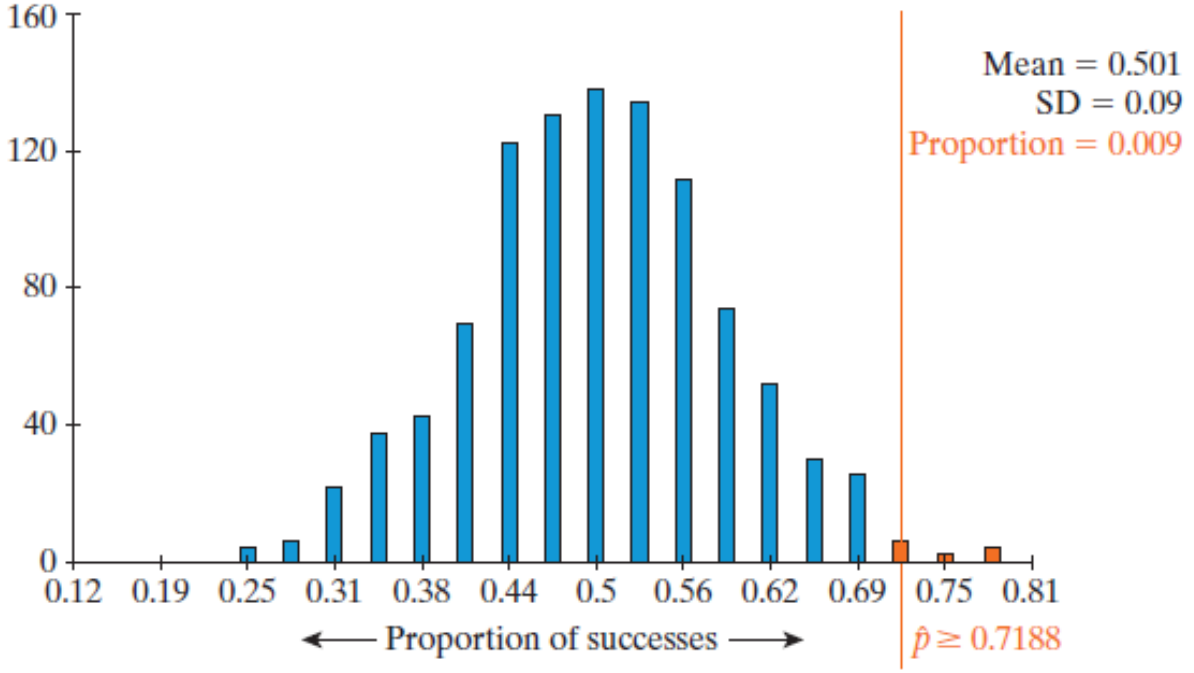
What is the *p*-value? What is the z-score?

$$z = \frac{(\text{observed statistic}) - (\text{mean of null distribution})}{(\text{SD of null distribution})}$$

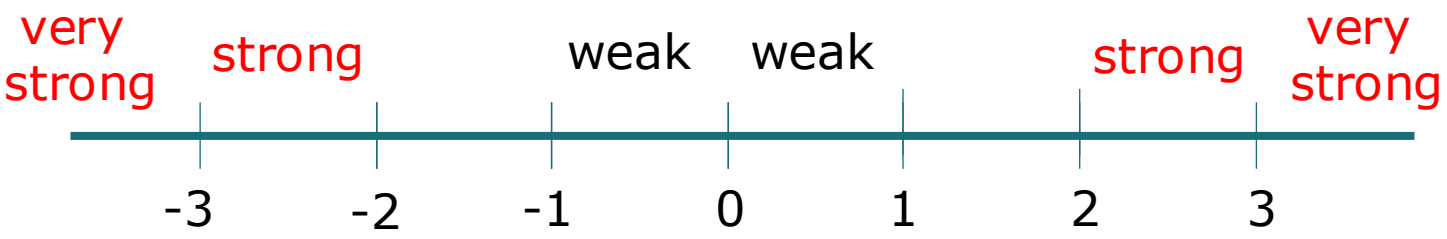


Is it good? – *p*-value

p-value	Evidence Against Null
above 0.10	not much
between 0.5 and 0.10	moderate
between 0.01 and 0.05	strong
below 0.01	very strong

Is it good? – standardized statistic

Standardized Statistic	Evidence Against Null
between −1.5 and 1.5	not much
below −1.5 or above 1.5	moderate
below −2 or above 2	strong
below −3 or above 3	very strong



Strength of evidence against H_0 for various values of z

Example 1.4: Predicting Elections (4 of 4)

- With a p-value of 0.009 we have strong evidence against the null hypothesis.
- When we calculate the standardized statistic, we again show strong evidence against the null.

$$z = \frac{0.7188 - 0.5}{0.09} = 2.43.$$

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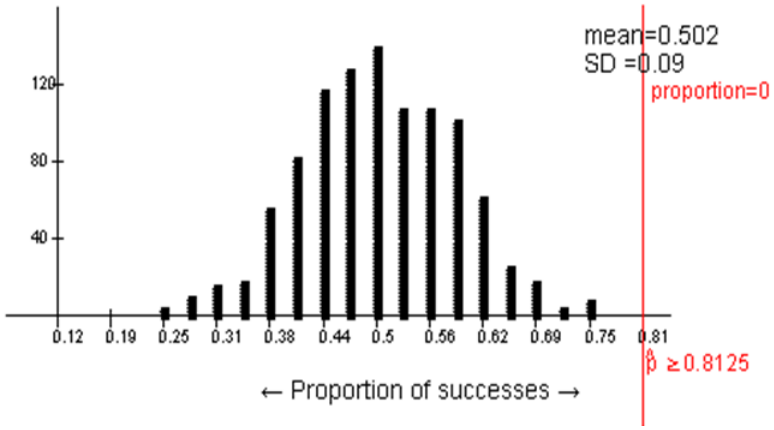
Example 1.4: Difference Between Observed Statistic and the Null Parameter (1 of 2)

- What if researchers correctly predicted 26 elections instead of 23?
  - $\frac{26}{32} = 0.8125$ never occurs just by chance

    hence the *p*-value is 0.



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Example 1.4: Difference Between Observed Statistic and the Null Parameter (2 of 2)

- The farther away the observed statistic is from the average value of the null distribution (or π_0), the more evidence there is against the null hypothesis.

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Example 1.4: What Affects the Strength of Evidence?

1. The difference between the observed statistic $(\hat{p})$ and null hypothesis parameter (π_0).
2. Sample size.
3. Whether we do a one or two-sided test.

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Example 1.4: Sample Size (1 of 4)

- Suppose the sample *proportion* stays the same, do you think increasing sample *size* will increase, decrease, or have no impact on the strength of evidence against the null hypothesis?

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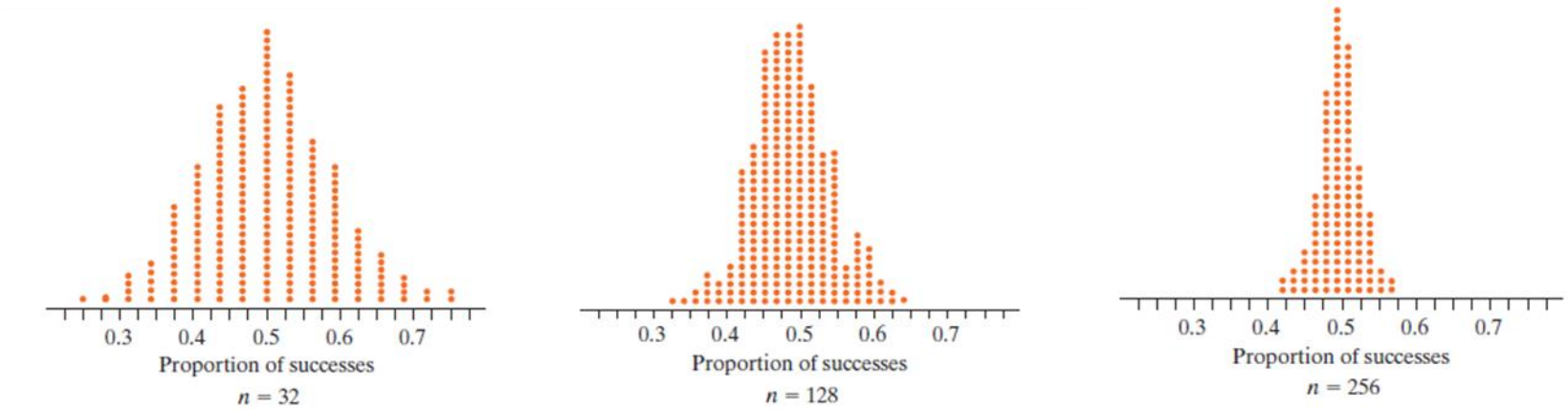
Example 1.4: Sample Size (2 of 4)

- The null distribution changes as we increase the sample size from 32 senate races to 128 races to 256 races.
- As the sample sizes increases, the variability (standard deviation) in the simulated statistics (sample proportions) decreases.



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Example 1.4: Sample Size (3 of 4)

- What does decreasing variability mean for statistical significance (with same sample proportion)?
- 32 elections
  - p-value = 0.009 and z = 2.43
- 128 elections
  - p-value = 0 and z =5.07
- 256 elections
  - Even stronger evidence
  - p-value = 0 and z = 9.52

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Reminder

p-value	Evidence Against Null
above 0.10	not much
between 0.5 and 0.10	moderate
between 0.01 and 0.05	strong
below 0.01	very strong

Standardized Statistic	Evidence Against Null
between −1.5 and 1.5	not much
below −1.5 or above 1.5	moderate
below −2 or above 2	strong
below −3 or above 3	very strong

Example 1.4: Sample Size (4 of 4)

- As the sample size increases, the variability in simulated sample proportions (the null distribution SD) decreases.
- Therefore, as the sample size increases, the evidence against the null hypothesis increases (*as long as the sample proportion stays the same* and is in the direction of the alternative hypothesis).

$$z = \frac{(\text{observed statistic}) - (\text{mean of null distribution})}{(\text{SD of null distribution})}$$

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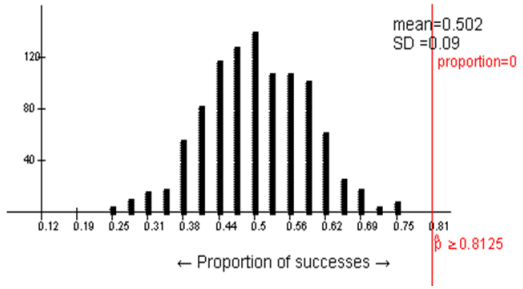
Example 1.4: Two-Sided Tests (1 of 5)

- What if researchers were wrong; instead of the person with the more competent face being elected more frequently, it was actually less frequently?

$$H_0: \pi = 0.5$$

$$H_a: \pi > 0.5$$

- With our original alternative, if we get a sample proportion less than 0.50, we would get a very large *p*-value.
- This is a **one-sided** test.
- Often one-sided is too restrictive.
- In fact most research uses two-sided tests.



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Example 1.4: Two-Sided Tests (2 of 5)

- In a two-sided test we have evidence in favor of the alternative hypothesis if the observed sample proportion is in either tail of the null distribution.

Null hypothesis: The probability this method predicts the winner equals 0.50. $\left(H_0: \pi = 0.50\right)$

Alternative hypothesis: The probability this method predicts the winner **is not** 0.50. $\left(H_a: \pi \neq 0.50\right)$

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Example 1.4: Two-Sided Tests (3 of 5)

- This change to the alternative hypothesis also affects how we compute the *p*-value.
- Remember that the *p*-value is the probability (assuming the null hypothesis is true) of obtaining a proportion that is equal to or **more extreme** than the observed statistic
- In a two-sided test, **more extreme** goes in both directions.

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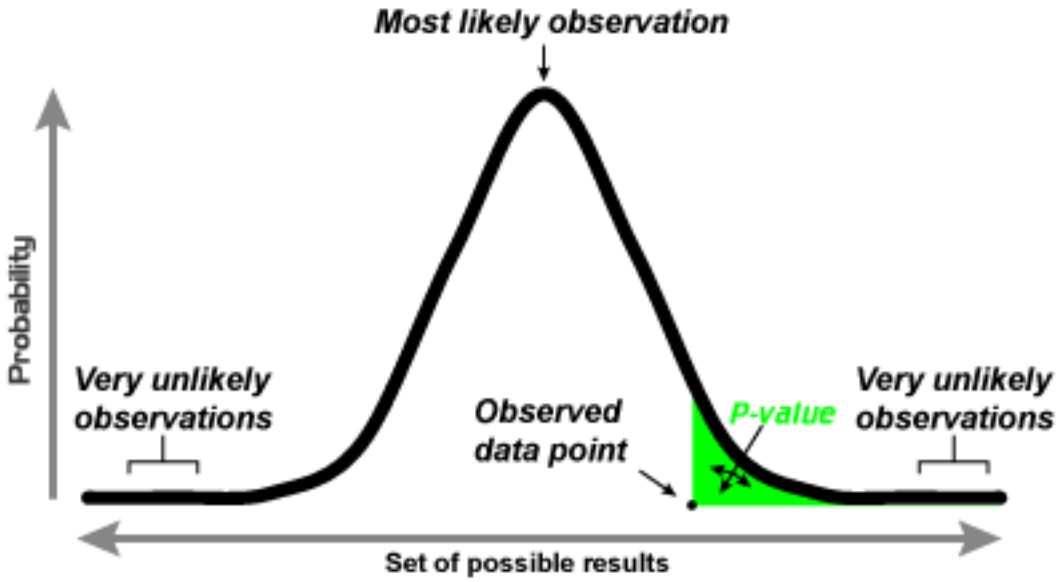
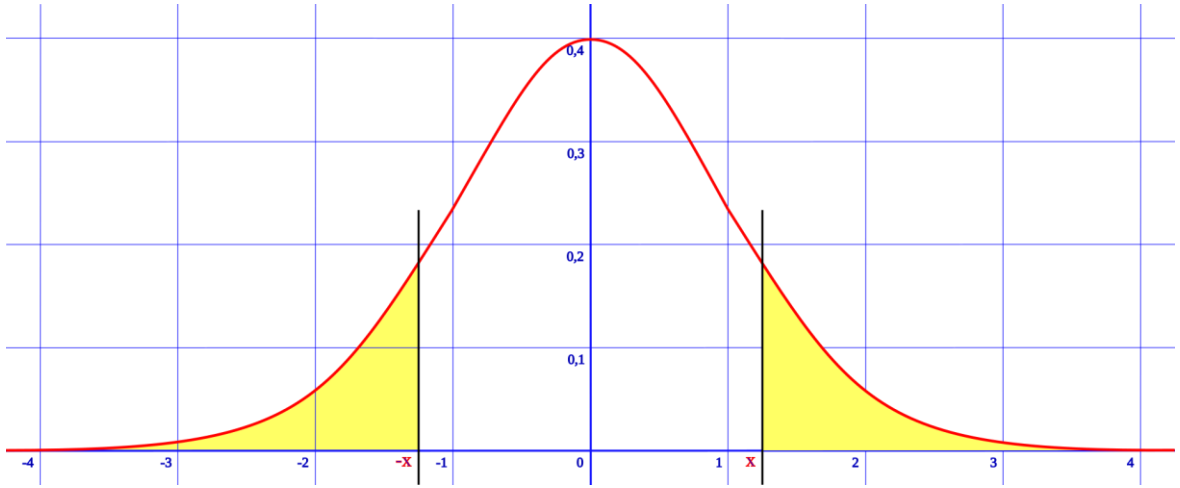
From Wikipedia



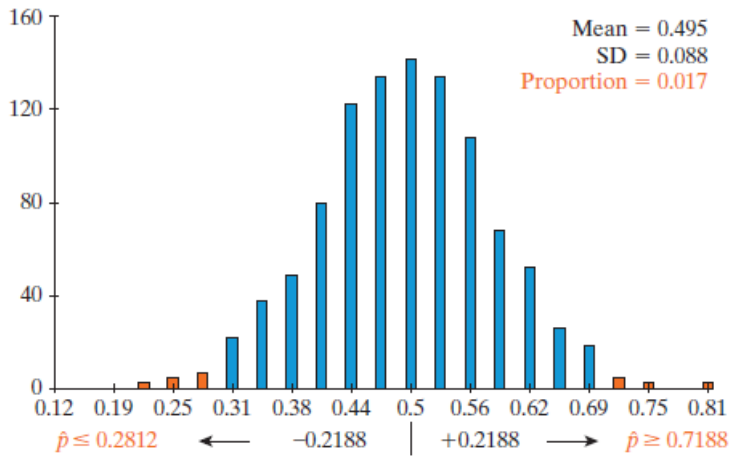
A p-value (shaded green area) is the probability of an observed (or more extreme) result arising by chance



https://en.wikipedia.org/wiki/One-_and_two-tailed_tests

Example 1.4: Two-Sided Tests (4 of 5)

- Because our sample proportion was 0.7188 and 0.7188 is 0.2188 above 0.5, we also need to look at 0.2188 below 0.50. (This gets a bit more complicated when the distribution is not symmetric, but the applet will do all the work for you.)
- The *p*-value will include all simulated proportions 0.7188 and above as well as those 0.2812 and below.



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Example 1.4: Two-Sided Tests (5 of 5)

- 0.7188 or greater was obtained 9 times
- 0.2812 or less was obtained 8 times
- The p-value is $\frac{(8+9=17)}{1000} = 0.017.$
- Two-sided tests increase the p-value (it about doubles) and hence decrease the strength of evidence.
- Two-sided tests are said to be more conservative. More evidence is needed to conclude alternative.

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Example 1.4: Predicting House Elections

(1 of 4)

- Researchers also predicted the 279 races for the House of Representatives in 2004.
- They correctly predicted the winner in $\frac{189}{279} \approx 0.677$, or 67.7% of the races.
- The House's sample percentage (67.7%) is bit smaller than the Senate's (71.9%), but the House's sample size is larger (279) than for the senate races (32).
- Do you expect the strength of evidence to be stronger, weaker, or essentially the same for the House compared to the Senate?

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Example 1.4: Predicting House Elections

(2 of 4)

Distance of the observed statistic to the null hypothesis value

- The observed statistic in the House is 0.677 compared to 0.719 in the Senate
- Slight decrease in the strength of evidence

Sample size

- The sample size is almost 10 times as large (279 vs. 32) in the House compared to the Senate
- This will increase the strength of evidence

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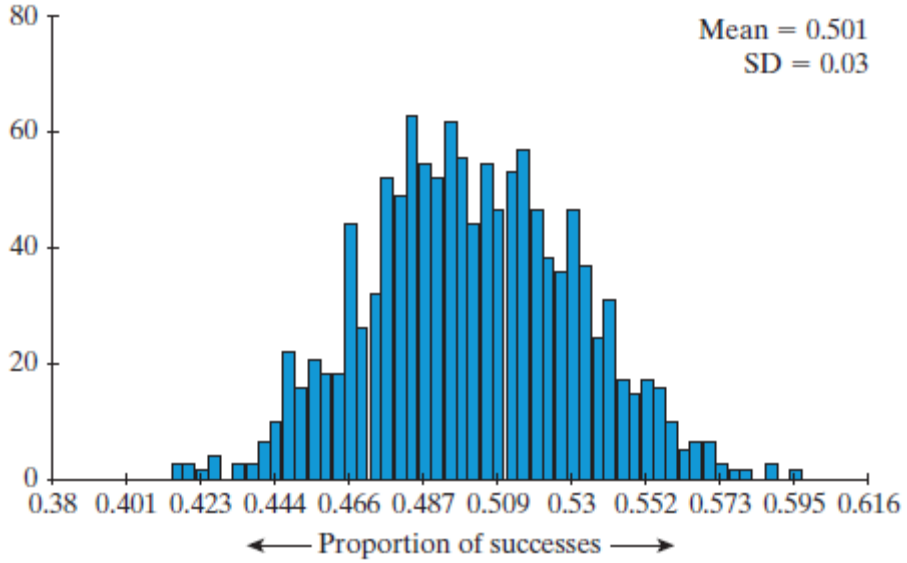
Example 1.4: Predicting House Elections

(3 of 4)

Null distribution of 279 sample House races



Simulated statistics ≥ 0.677 didn't occur hence the *p*-value is approximately 0

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Example 1.4: Predicting House Elections

(4 of 4)

- What about the standardized statistics?
  - For the Senate it was 2.43
  - For the House it is 5.90.
- The larger sample size for the House trumped its smaller proportion, so we have stronger evidence against the null using the data from the House.

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